

VECM Summary from RW4 Primary Chain

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end. Retail_combined is the anchored downstream index built from Silpo effective prices, Novus observed prices, and a level-aligned ConsumerUA anchor, while Retail_combined_core keeps the strict retailer-only overlap for robustness.

family=VECM

rows=288

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk_Dairy_Research/outputs/primary_chain_summary/primary_chain_consolidated.xlsx

Table: VECM_Summary

panel_level	panel_name	segment_name	product	standardized_type	retailer_panel	brand	price_variant	reconstruction_variant	farm_gate_source	frequency	intersection_rule	pair_scope	chain_direction	stage_from	stage_to	link_model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	unreliable_reason	shock_dummy_included	promo_controls_included	ljungbox_p	arch_p	jb_p	notes	robust_linear_vs_pchip	interpolation_sensitive	robust_across_reconstruction	core_finding_flag
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	220	nan	nan	0.2824524537101993	nan	nan	3.0	ok	0	nan	0	0.6294193986152101	0.5849111109108772	1.710159696936651e-23	k_ar_diff=5	0	1	0		
pairwise_product	pairwise::producer_prozor...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	216	nan	nan	-0.02283809002587641	nan	nan	3.0	ok	0	nan	0	0.5597426038982174	0.6782497405568844	0.0	k_ar_diff=1	0	1	0		
pairwise_product	pairwise::producer_prozor...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	189	nan	nan	-0.0712503149399635	nan	nan	3.0	ok	0	nan	0	0.5048040691862214	0.5800919411794552	0.001240756648566582	k_ar_diff=1	0	1	0		
pairwise_product	pairwise::producer_prozor...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	178	nan	nan	0.4779763939385675	nan	nan	3.0	unreliable	1	residual_autocorr_or_arch	0	0.03832434425564643	0.001405090704417639	0.006521864031314946	k_ar_diff=1	0	1	0		
product	product::Вершки::Silpo::o...	Вершки	Вершки	cream	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.214812140355627	nan	nan	2.0	ok	0	nan	0	0.9397102432548599	0.785228977301169	0.02977676690767751	k_ar_diff=5	0	1	0		
product	product::Молоко питне::Si...	Молоко питне	Молоко питне	milk	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.542086532312744	nan	nan	2.0	ok	0	nan	0	0.9910630862290195	0.9998696776320517	3.549057433899505e-160	k_ar_diff=5	0	1	0		
product	product::Сир твердий::Si...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-3.281200511143566	nan	nan	3.0	ok	0	nan	0	0.9908513112556221	0.5717024280852399	0.1445401016754244	k_ar_diff=5	0	1	0		
product	product::Сметана::Silpo::...	Сметана	Сметана	sour_cream	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-24.70966978176242	nan	nan	2.0	ok	0	nan	0	0.1646681594712951	0.9828120691897677	0.9191199356810634	k_ar_diff=5	0	1	0		
product	product::Вершки::b...	Вершки	Вершки	cream	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.214812140355627	nan	nan	2.0	ok	0	nan	0	0.9397102432548599	0.785228977301169	0.02977676690767751	k_ar_diff=5	0	1	0		
product	product::Молоко питне::Si...	Молоко питне	Молоко питне	milk	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	41	nan	nan	2.542086532312744	nan	nan	2.0	ok	0	nan	0	0.9910630862290195	0.9998696776320517	3.549057433899505e-160	k_ar_diff=5	0	1	0		
product	product::Сир твердий::Si...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-3.281200511143566	nan	nan	3.0	ok	0	nan	0	0.9908513112556221	0.5717024280852399	0.1445401016754244	k_ar_diff=5	0	1	0		
product	product::Сметана::Silpo::...	Сметана	Сметана	sour_cream	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	38	nan	nan	-24.70966978176242	nan	nan	2.0	ok	0	nan	0	0.1646681594712951	0.9828120691897677	0.9191199356810634	k_ar_diff=5	0	1	0		
product	product::Вершки::Retail_c...	Вершки	Вершки	cream	Retail_combined	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	48	nan	nan	2.849247528755007	nan	nan	1.0	ok	0	nan	0	0.979829496472063	0.9074538305651549	6.42611842786754e-08	k_ar_diff=1	0	1	0		
product	product::Молоко питне::Re...	Молоко питне	Молоко питне	milk	Retail_combined	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	216	nan	nan	-0.02283809002587641	nan	nan	3.0	ok	0	nan	0	0.5597426038982174	0.6782497405568844	0.0	k_ar_diff=1	0	1	0		
product	product::Сир твердий::Ret...	Сир твердий	Сир твердий	hard_cheese	Retail_combined	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	189	nan	nan	-0.0712503149399635	nan	nan	3.0	ok	0	nan	0	0.5048040691862214	0.5800919411794552	0.001240756648566582	k_ar_diff=1	0	1	0		
product	product::Сметана::Retail_...	Сметана	Сметана	sour_cream	Retail_combined	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	178	nan	nan	0.4779763939385675	nan	nan	3.0	unreliable	1	residual_autocorr_or_arch	0	0.03832434425564643	0.001405090704417639	0.006521864031314946	k_ar_diff=1	0	1	0		
product	product::Вершки::Retail_c...	Вершки	Вершки	cream	Retail_combined	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	48	nan	nan	2.849247528755007	nan	nan	1.0	ok	0	nan	0	0.979829496472063	0.9074538305651549	6.42611842786754e-08	k_ar_diff=1	0	1	0		
product	product::Молоко питне::Re...	Молоко питне	Молоко питне	milk	Retail_combined	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	216	nan	nan	-0.02283809002587641	nan	nan	3.0	ok	0	nan	0	0.5597426038982174	0.6782497405568844	0.0	k_ar_diff=1	0	1	0		
product	product::Сир твердий::Ret...	Сир твердий	Сир твердий	hard_cheese	Retail_combined	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	189	nan	nan	-0.0712503149399635	nan	nan	3.0	ok	0	nan	0	0.5048040691862214	0.5800919411794552	0.001240756648566582	k_ar_diff=1	0	1	0		
product	product::Сметана::Retail_...	Сметана	Сметана	sour_cream	Retail_combined	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	178	nan	nan	0.4779763939385675	nan	nan	3.0	unreliable	1	residual_autocorr_or_arch	0	0.03832434425564643	0.001405090704417639	0.006521864031314946	k_ar_diff=1	0	1	0		
product	product::Вершки::Retail_c...	Вершки	Вершки	cream	Retail_combined_core	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	48	nan	nan	2.849247528755007	nan	nan	1.0	ok	0	nan	0	0.979829496472063	0.9074538305651549	6.42611842786754e-08	k_ar_diff=1	0	1	0		
product	product::Молоко питне::Re...	Молоко питне	Молоко питне	milk	Retail_combined_core	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	50	nan	nan	-6.574215974517948	nan	nan	1.0	ok	0	nan	0	0.9772225899789795	0.9999804223319424	1.373585134027023e-290	k_ar_diff=5	0	1	0		
product	product::Сир твердий::Ret...	Сир твердий	Сир твердий	hard_cheese	Retail_combined_core	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	46	nan	nan	2.348309998847764	nan	nan	2.0	ok	0	nan	0	0.6093519858451928	0.7207783449280616	k_ar_diff=5	0	1	0			
product	product::Сметана::Retail_...	Сметана	Сметана	sour_cream	Retail_combined_core	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	42	nan	nan	-13.61589768417084	nan	nan	1.0	ok	0	nan	0	0.7781781013550557	0.3716790279134638	0.5768903998519131	k_ar_diff=5	0	1	0		
product	product::Вершки::Retail_c...	Вершки	Вершки	cream	Retail_combined_core	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	48	nan	nan	2.849247528755007	nan	nan	1.0	ok	0	nan	0	0.979829496472063	0.9074538305651549	6.42611842786754e-08	k_ar_diff=1	0	1	0		
product	product::Молоко питне::Re...	Молоко питне	Молоко питне	milk	Retail_combined_core	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	50	nan	nan	-6.574215974517948	nan	nan	1.0	ok	0	nan	0	0.9772225899789795	0.9999804223319424	1.373585134027023e-290	k_ar_diff=5	0	1	0		
product	product::Сир твердий::Ret...	Сир твердий	Сир твердий	hard_cheese	Retail_combined_core	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	46	nan	nan	2.348309998847764	nan	nan	2.0	ok	0	nan	0	0.6093519858451928	0.7207783449280616	k_ar_diff=5	0	1	0			
product	product::Сметана::Retail_...	Сметана	Сметана	sour_cream	Retail_combined_core	nan	baseline	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	42	nan	nan	-13.61589768417084	nan	nan	1.0	ok	0	nan	0	0.7781781013550557	0.3716790279134638	0.5768903998519131	k_ar_diff=5	0	1	0		
average	average::Silpo::observed:... all_products	average	average	all_products	average	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	average	average	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	47	nan	nan	1.0	ok	0	nan	0	0.2270834451270314	0.6453889741668349	2.800874446122529e-10	k_ar_diff=5	0	1	0		
average	average::Silpo::observed:... all_products	average	average	all_products	average	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	average	average	forward	ProducerUA	Retail	producer_prozorro_retail...	VECM	retail	producer+prozorro	47	nan	nan	1.0	ok	0	nan	0	0.3968749311479061	0.5363568606653772	k_ar_diff=1	0	1	0			
average	average::Silpo::observed:... all_products	average	average	all_products	average	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	average	average	forward	FarmGateUA	Retail	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	47	nan	nan	2.0	ok	0	nan	0	-0.02072783869708888	nan	nan	nan	2.0	ok	0		
average	average::Silpo::baseline:... all_products	average	average	all_products	average	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	average	average	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	47	nan	nan	1.0	ok	0	nan	0	-26.52767975899912	0.6453889741668349	2.800874446122529e-10	k_ar_diff=5	0	1	0		
average	average::Silpo::baseline:... all_products	average	average	all_products	average	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	average	average	forward	ProducerUA	Retail	producer_prozorro_retail...	VECM	retail	producer+prozorro	47	nan	nan	1.0	ok	0	nan	0	0.3968749311479061	0.5363568606653772	k_ar_diff=1	0	1	0			
average	average::Silpo::baseline:... all_products	average	average	all_products	average	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	average	average	forward	FarmGateUA	Retail	farmgate_producer_prozorrr...	VECM	retail	farm_gate+producer+prozorro	47	nan	nan	2.0	ok	0	nan	0	-0.02072783869708888	nan	nan	nan	2.0	ok	0		
average	average::Silpo::baseline:... all_products	average	average	all_products	average	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	average	average	forward	FarmGateUA	ProZorro	farmgate_producer_prozorrr...	VECM	stage_3	stage_1+stage_2	47	nan	nan	1.0	ok	0	nan	0	-26.52767975899912	0.6453889741668349	2.800874446122529e-10	k_ar_diff=5	0	1	0		
average	average::Silpo::baseline:... all_products	average	average	all_products	average	Silpo	nan	baseline	linear	initial	daily	chain_common_support	full_chain	average	average	forward	ProducerUA	Retail	producer_prozorro_retail...	VECM	retail	producer+prozorro	47	nan	nan	1.0	ok	0	nan	0	0.3968749311479061	0.5363568606653772	k_ar_diff=1	0	1	0			
average	average::Silpo::baseline:... all_products																																						